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工作经历

博士后研究员 2023.10 - 至今

波士顿大学电子与计算机工程系 *Ji-xin Cheng* 教授课题组 波士顿

特聘副研究员 2021.7 - 2023.10

北京大学物理学院施可彬教授课题组 北京

博士后 2019.7 - 2021.6

北京大学物理学院施可彬教授课题组 北京

教育背景

北京大学 博士-光学 导师: 施可彬教授 2014 - 2019

南开大学 本科-物理学 2010 - 2014

专业技能

[光学] 研究方向：计算成像、非线性光学、全息/数字全息、显微成像、光谱学

技术专长：复杂光学成像系统的方案设计、光路搭建与调试、系统稳定性与性能优化

应用领域：活细胞成像、无标记/荧光成像、多模态与超分辨显微成像

[编程] MATLAB, C/C++, Labview, Python, Bash, LaTeX

- **仪器控制与自动化**：精通图像/信号采集与控制程序开发，熟练掌握嵌入式系统软硬件协同设计
- **图像/信号处理**：熟练掌握常用机器视觉和图像/信号处理算法，具有丰富的相关算法开发与优化经验
- **高新能计算**：掌握基于 *GPU* 与 *Linux* 集群的并行计算编程及实时图像处理程序开发

[软件] Fiji/ImageJ, Mathematica, OriginPro, Adobe AI/PS, Zemax, COMSOL

代表性论文

(1) **Dashan Dong**#, *Danchen Jia, Xinyan Teng, Jianpeng Ao, George Abu-Aqil, Biwen Gao, Meng Zhang, Qing Xia, Ji-Xin Cheng*✉. " Chem-SIM: Super-resolution Chemical Imaging via Photothermal Modulation of Structured-Illumination Fluorescence ." *arXiv*, (2026).

(2) *Lige Liu*#, **Dashan Dong**#, *Zhiwei Long, Wanxue Wei, Chang Sun, Wei Liu, Xiaoshuai Huang, Liangyi Chen, Haizheng Zhong*✉, *and Kebin Shi*✉. " Single particle fluorescence imaging of perovskite nanocrystal crystallization for illustrating coupled nucleation-and-growth ." *Nature Communications*, 16.5664, (2025).

(3) *Shuqi Mu*#, *Yingtong Shi, Yintong Song, Wei Liu, Wanxue Wei, Qihuang Gong,* **Dashan Dong**✉, *and Kebin Shi.* " Multislice computational model for birefringent scattering ." *Optica*, 10.1, (2023).

(4) *Shuqi Mu*#, *Yu Yang, Juhao Li,* ***Dashan Dong***✉, *Ruijun Lan*✉, *Kebin Shi.* " Three dimension refractive index characterization for photonic waveguides ." *Journal of Lightwave Technology*, 40.8, (2022).

(5) **Dashan Dong**#, *Xiaoshuai Huang*#, *Liuju Li*#, *Heng Mao, Yanquan Mo, Guangyi Zhang, Zhe Zhang, Jaiyu Shen, Wei Liu, Zeming Wu, Guanghui Liu, Yanmei Liu, Hong Yang, Qihuang Gong, Kebin Shi*✉, *and Liangyi Chen*✉. " Super-resolution fluorescence-assisted diffraction computational tomography reveals the three-dimensional landscape of the cellular organelle interactome ." *Light: Science & Applications*, 9.11, (2020).

其它学术论文

(1) *Tongyu Li*[#], *Yi Shen*, **Dashan Dong**, *Danchen Jia, Jianpeng Ao, Ji-Xin Cheng, Lei Tian*[✉]. " Transfer-function approach to substrate-enhanced diffraction tomography ." *Optica*, 13.4, (2026).

(2) *George Abu-Aqil*[#], **Dashan Dong**[#], *Jiaze Yin*[#], *Jianpeng Ao, Hongjian He, Guangrui Ding, Qing Xia, Ji-Xin Cheng*[✉]. " Bond-Selective Imaging via Vibrational Relaxation Encoded Fluorescence ." *The Journal of Physical Chemistry Letters*, 16, (2025).

(3) *Jia Danchen*[#], *Huang Steven*[#], *Tulegenov Dias*, **Dashan Dong**, *Shvets Gennady*[✉], *Cheng Ji-Xin*[✉]. " Metasurface-enhanced infrared photothermal microscopy towards ultrasensitive chemical analysis ." *Advanced Photonics*, 7.5, (2025).

(4) *Qian Chen*[#], *Wen Gou*[#], *Wenqing Lu*[#], *Jie Li, Yuhong Wei, Haoyu Li, Chengyu Wang, Wei You, Zhengqian Li*, **Dashan Dong**, *Xiuli Bi, Bin Xiao*[✉], *Liangyi Chen*[✉], *Kebin Shi*[✉], *Junchao Fan*[✉], *Xiaoshuai Huang*[✉]. " Fast, three-dimensional, live-cell super-resolution imaging with multiplane structured illumination microscopy ." *Nature Photonics*, (2025).

(5) *Jiaze Yin*[#], *Christian Pfluegl, Chu C. Teng, Rylie Bolarinho, Guo Chen, Xinrui Gong*, **Dashan Dong**, *Daryoosh Vakhshoori, Ji-Xin Cheng*[✉]. " Mid-infrared Energy Deposition Spectroscopy ." *Physical Review Letters*, 134.9, (2025).

(6) *Yue Wang*[#], *Jingrui Gong, Ning Xu, Shaohui Yan*, **Dashan Dong**, *Kebin Shi*[✉]. " Large Field of View and Isotropic Light Sheet Microscopy with Aberration-Free Tunable Foci ." *Laser & Photonics Reviews*, 19.3, (2025).

(7) *Wenkai Yang*[#], *Zijian Wang, Jian Xu*, **Dashan Dong**, *Guiyuan Cao, Han Lin, Baohua Jia, Lige Liu*[✉], *Kebin Shi*[✉]. " Ultracompact computational spectroscopy with a detour-phased planar lens ." *Light: Advanced Manufacturing*, 5.44, (2024).

(8) *Ziheng Ji*[#], *Wentao Yu*, **Dashan Dong**, *Hong Yang, Kaihui Liu, Yun-Feng Xiao, Qihuang Gong, Qinghai Song*[✉], *Kebin Shi*[✉]. " High spatial resolution collinear chiral sum-frequency generation microscopy ." *Advanced Photonics Nexus*, 3.2, (2024).

(9) *Yue Wang*[#], **Dashan Dong**, *Wenkai Yang, Renxi He, Ming Lei, and Kebin Shi*[✉]. " Reflective ultrathin light-sheet microscopy with isotropic 3D resolutions ." *Photonics Research*, 12.2, (2024).

(10) *Peng Liu*[#], *Jing Shi*[#], *Danli Sheng*[#], *Wenqing Lu, Jie Guo, Lei Gao, Xiaoqing Wang, Shaofeng Wu, Yanwen Feng*, **Dashan Dong**, *Xiaoshuai Huang*[✉], *and Hongyun Tang*[✉]. " Mitopherogenesis, a form of mitochondria-specific ectocytosis, regulates sperm mitochondrial quantity and fertility ." *Nature Cell Biology*, 25, (2023).

(11) 魏婉雪[#], 何苗, 徐坚, **董大山**[✉]. " 一种大视野快速光学衍射层析成像技术 ." 中国激光, 50.3, (2023).

(12) *Wenkai Yang*[#], *Lige Liu*[✉], **Dashan Dong**, *Yunan Gao, Hong Yang, Qihuang Gong, and Kebin Shi*[✉]. " *In situ* three-dimensional observation of perovskite crystallization revealed by two-photon fluorescence imaging ." *Advanced Optical Materials*, 10.13, (2022).

(13) *Wenkai Yang*[#], *Lige Liu*[#], **Dashan Dong**, *Xin Zhang, Han Lin, Yunkun Wang, Hong Yang, Yunan Gao, Haizheng Zhong, Baohua Jia, and Kebin Shi*[✉]. " Detour-phased perovskite ultrathin planar lens using direct femtosecond laser writing ." *Photonics Research*, 10.12, (2022).

(14) *Runfeng Li*[#], *Ruijun Lan*[✉], **Dashan Dong**, *Hong Yang, and Kebin Shi*[✉]. " Bessel beam coherent anti-Stokes Raman scattering spectroscopy for turbulent flow diagnosis ." *Applied Spectroscopy*, (2022).

(15) 穆书奇[#], **董大山**, 施可彬[✉]. " 无标记光学成像技术 ." 激光与光电子学进展, 59.12, (2022).

(16) 李润丰[#], **董大山**, 施可彬[✉]. " 光场调控在相干拉曼散射光谱与成像中的应用 (特邀) ." 光子学报, 50.1, (2022).

(17) *Wei Liu*[#], **Dashan Dong**, *Hong Yang, Qihuang Gong, and Kebin Shi*[✉]. " Robust and high-speed rotation control in optical tweezers by using polarization synthesis based on heterodyne interference ." *Opto-Electronic Advances*, 3.8, (2020).

(18) **Dashan Dong**[#], *Kebin Shi*[✉]. " Solving the missing cone problem by deep learning ." *Advanced Photonics*, 2.2, (2020).

(19) *Shuanglong Liu*[#], *Bowen Sheng, Xinqiang Wang*[✉], **Dashan Dong**, *Ping Wang, Zhaoying Chen, Tao Wang, Xin Rong, Duo Li, Liuyun Yang, Shangfeng Liu, Mo Li, Jian Zhang, Weikun Ge, Kebin Shi, Yuzhen Tong, Bo Shen.* " Molecular beam epitaxy of single-crystalline aluminum film for low threshold ultraviolet plasmonic nanolasers ." *Applied Physics Letters*, 122.23,

(2018).

(20) *Wentao Yu#, Ziheng Ji,* **Dashan Dong**, *Xusan Yang, Yunfeng Xiao, Qihuang Gong, Peng Xi✉, and Kebin Shi✉.* " Superresolution deep imaging with hollow Bessel beam STED microscopy ." *Laser & Photonics Reviews*, 10.1, (2016).

(21) *Yonggang Lv#, Ziheng Ji,* **Dashan Dong**, *Kebin Shi✉, and Qihuang Gong.* " Wide-field vibrational phase contrast imaging based on coherent anti-Stokes Raman scattering holography ." *Chinese Physics Letters*, 32.7, (2015).

(22) *Bin Tsai#, Wei Liu,* **Dashan Dong**, *Kebin Shi, Liangyi Chen, Ning Gao✉.* " Phase separation of Mer2 organizes the meiotic loop-axis structure of chromatin during meiosis I ." *bioRXiv*, (2020).

会议论文

(1) **Dashan Dong** *and Ji-Xin Cheng.* " Super-resolution Chemical Imaging via Structured Illumination Fluorescence-Detected Mid-Infrared Photothermal Microscopy ." *Novel Techniques in Microscopy*, Optica Biophotonics Congress NTu3C.3 (2025).

(2) *Wanxue Wei,* **Dashan Dong**, *Kebin Shi.* " Fast optical diffraction tomography microscopy with large field of view and lossless pupil beam combination ." *Ultrafast Nonlinear Imaging and Spectroscopy XII*, Proc. SPIE 1313910, (2024).

(3) **Dashan Dong**, *Xiaoshuai Huang, Liuju Li, Kebin Shi, and Liangyi Chen.* " Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome ." *Advanced Optical Imaging Technologies III*, SPIE/COS Photonics Asia 115490G, (2020).

(4) **Dashan Dong**, *Xiaoshuai Huang, Liuju Li, Kebin Shi, and Liangyi Chen.* " Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome ." *Digital Holography and Three-Dimensional Imaging 2020*, Imaging and Applied Optics Congress, Optica Publishing Group HF1G.6, (2020).

(5) **Dashan Dong**, *Yanhui Cai, Ziheng ji, Hong Yang, Qihuang Gong, and Kebin Shi.* " Tomographic Diffractive Microscopy for Better 3D Imaging ." *14th International Conference on Photonics and Imaging in Biology and Medicine*, Optica Publishing Group W3A.43, (2017).

(6) *Kebin Shi,* **Dashan Dong**, *Yanhui Cai, Wei Liu, Chendi Shao.* " High resolution nonlinear imaging based on optical field engineering ." *Ultrafast Nonlinear Imaging and Spectroscopy V*, Proc. SPIE 1038000Z, (2017).

专利

(1)	一种双模态显微成像系统和方法	国际专利 PCT/CN2021/071393, (2020)
(2)	一种基于缺失型反射镜的离轴全息合束装置及其方法	2023110715739, (2023)
(3)	一种双模态显微成像系统及其成像方法	ZL202110946936.3, (2022)
(4)	基于微棱镜的反射式轴向光片荧光显微成像装置及方法	ZL 202110817118.3, (2022)
(5)	一种基于贝塞尔 CARS 的湍流光谱测量系统及其测量方法	ZL202110667298.1, (2022)
(6)	一种双模态显微成像系统和方法	ZL202010059510.1, (2022)
(7)	基于光学外差干涉法的动态柱矢量光场产生装置及其方法	ZL202010493775.2, (2021)
(8)	一种基于 GB-STED 的深层超分辨激光直写系统及其实现方法	ZL202010069141.4, (2020)

科研项目

[课题负责人] 国家重点研发计划下设青年科学家课题 (2022YFC3401103) 2022

活细胞超分辨率三维全景成像与解析技术体系研发课题：无标记成像与大数据机器学习 234 万元

[主持] 国家自然科学基金青年科学基金项目 (12004013) 2021

用于活细胞成像的反射式光学衍射层析成像及其应用 24 万元

[共同负责人] 北京大学临床医学 +X 青年专项 2022

基于无标记成像探究脂滴在卵不对称分裂中的作用 10 万元

[主持] 中国博士后科学基金第 68 批面上资助 (2020M680220) 2021

超分辨荧光辅助的光学衍射层析成像及其应用 8 万元

[参与] 国家自然科学基金面上项目 (12174010) 2022

微腔混沌动力学的超高速光学成像 62 万元

会议报告

[口头报告] Optica Biophotonics Congress 2025 2025.4

Super-Resolution Chemical Imaging via Structured Illumination Fluorescence-Detected Mid-Infrared Photothermal Microscopy 美国-圣地亚哥

[墙报] Chemical Imaging 2023, Gordon Research Conference 2023.8

Optical Diffraction Tomography Reveals the 3D Landscape of Living Cells 美国-波士顿

[邀请报告] 第一届先进成像与信息处理会议暨 2023 中国光学学会全息与光信息处理专委会学术年会 2023.7

面向低光毒性的生物光学成像新技术 江西-井冈山

[邀请报告] 第三届全国光子技术论坛 2023.4

基于光场调控的高分辨成像及应用 广东-广州

[邀请报告] 第二届国际计算成像会议 (CITA2022) 2022.10

基于光学衍射层析的活细胞全景成像 上海 (线上)

[邀请报告] 第五届光学青年科学家论坛 (OYSS2022) 2022.9

基于光场调控的高分辨成像及应用 福建-福州

[邀请报告] 第二十届全国基础光学与光物理学术讨论会 2021.9

基于四维时空连续性的光学衍射层析成像 吉林-延吉

[邀请报告] 2021 年中国光学学会学术大会 2021.9

基于四维时空连续性的光学衍射层析成像 广东-深圳

[邀请报告] 第四届光学青年科学家论坛 (OYSS2020) 2020.12

光学衍射层析显微成像及其应用 浙江-宁波

[口头报告] 第二届全国光子技术论坛 2020.11

光学衍射层析显微成像及其应用 广东-广州

[邀请报告] 2020 年全国电子显微学学术年会 2020.11

基于光学衍射层析的活细胞全景成像 四川-成都

[口头报告] 第十二届全国光学青年学术论坛 2020.11

光学衍射层析显微成像及其应用 河北-保定

[口头报告] 2020 年亚洲光电子会议 (Photonics Asia 2020) 2020.10

Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome 线上

[口头报告] Digital Holography and Three-Dimensional Imaging, Imaging and Applied Optics Congress 2020.6

Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome 线上

[墙报] International Conference on Photonics and Imaging in Biology and Medicine 2017.9

Tomographic Diffractive Microscopy for Better 3D Imaging 江苏-苏州

[墙报] SPIE Optics + Photonics 2017 2017.8

Tomographic Diffractive Microscopy for Better 3D Imaging 美国-圣迭戈

[口头报告] 中国物理学会 2016 年秋季学术会议 2016.8

光学衍射断层显微成像 北京

学术服务

[审稿人] Photonics Research, Optics Letters, Optics Express, Advanced Science, APL Photonics, 中国激光, 光学学报

[程序委员] 国际计算成像会议 (CSOE-CITA), 专题 9: 生物医学与计算成像 2022, 2023

[青年编委] 《中国激光》杂志 2022, 2023

[志愿者] SPIE/COS 亚洲光电子会议 2016, 2022

[主席] 北京大学 SPIE 学生分会 2017

[副主席] 北京大学 SPIE 学生分会 2016

学位论文

[博士学位论文] 光学衍射层析显微成像的研究 导师: 施可彬教授, 北京大学

[本科毕业论文] 新型矢量光场的特性研究 指导教师: 李勇男教授, 南开大学

荣誉奖项

[2022] 人工微结构和介观物理国家重点实验室甘子钊优秀博士后奖 北京大学

[2022] 现代光学所重要科研进展 北京大学

[2017] 五四奖学金 北京大学

[2017] 优秀科研单项奖 北京大学

[2013] 合展励学金 中国科技馆发展基金会

[2012] 国家励志奖学金 南开大学

[2012] 第三届中国大学生物理学术竞赛-团队一等奖

[2012] 三好学生 南开大学

[2011] 优秀学生基础学科奖学金 南开大学